

8.1 Stationary Waves

Question Paper

Course	CIEA Level Physics
Section	8. Superposition
Topic	8.1 Stationary Waves
Difficulty	Medium

Time allowed: 20

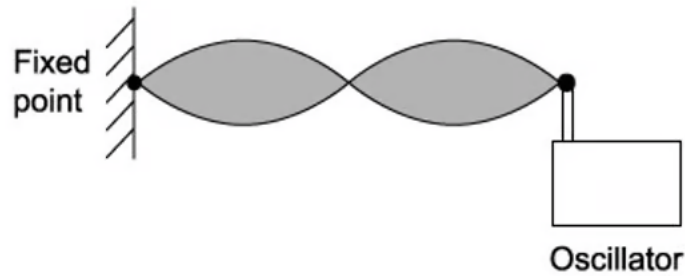
Score: /10

Percentage: /100

Question 1

A student was investigating the speed of a transverse wave on a stretched spring. They found that by adjusting the tension on the spring, they were able to change the speed.

The wave pattern produced is shown in the diagram the frequency of the oscillator was set to 650 Hz



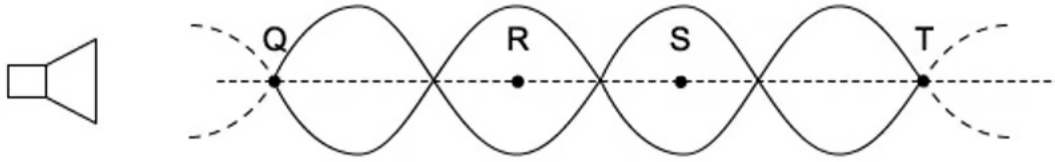
What needs to be changed to maintain the same wave pattern when the frequency is increased to 750 Hz?

- A. increase the speed of the wave on the string
- B. increase the wavelength of the wave on the string
- C. decrease the wavelength of the wave on the string
- D. decrease the speed of the wave on the string

[1 mark]

Question 2

The pattern produced when two waves are produced by superposition of sound waves from a loudspeaker reflected by a metal sheet.



Q, R, S and T are four points on the line through the centre of these waves.

Which of the following statements is correct?

- A. a node is a quarter of a wavelength from an adjacent antinode
- B. the stationary waves oscillate at right angles to the line QT
- C. an antinode is formed at the surface of the metal sheet
- D. the oscillations at R are in phase with those at S

[1 mark]

Question 3

Headphones that have noise reduction can cancel out external noise by producing their own waves.

They are fitted with a microphone that detects the external sound frequency. The loudspeaker then produces a wave with the same frequency but different phase.

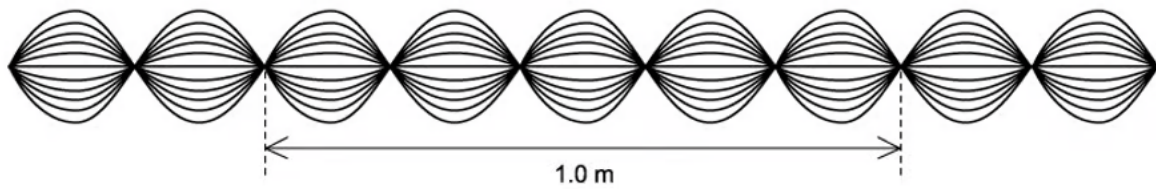
What would the phase difference needed to cancel out the external sound?

- A. 360°
- B. 270°
- C. 180°
- D. 90°

[1 mark]

Question 4

The diagram below shows a wave pattern over time



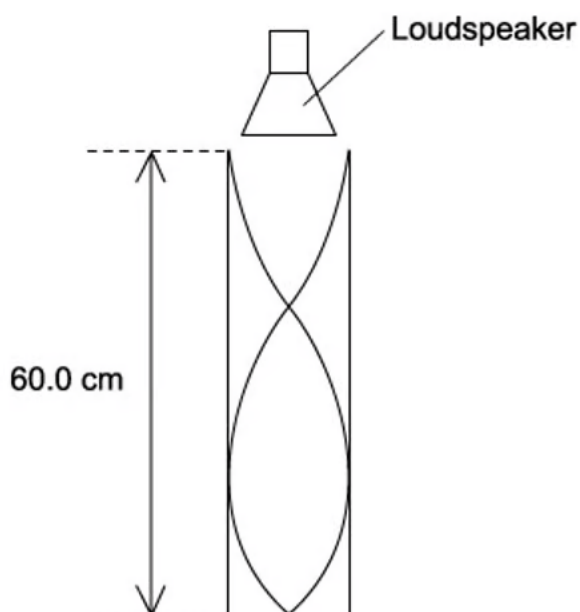
What is the correct description of the wave?

- A. the wave is transverse, has a wavelength of 40 cm and is stationary
- B. the wave is transverse, has a wavelength of 40 cm and is progressive
- C. the wave is transverse, has a wavelength of 20 cm and is stationary
- D. the wave is longitudinal, has a wavelength of 20 cm and is stationary

[1 mark]

Question 5

The diagram below shows sound wave produced in a column by a loudspeaker. The speed of sound in air is 330 m s^{-1}



What is the frequency of the wave?

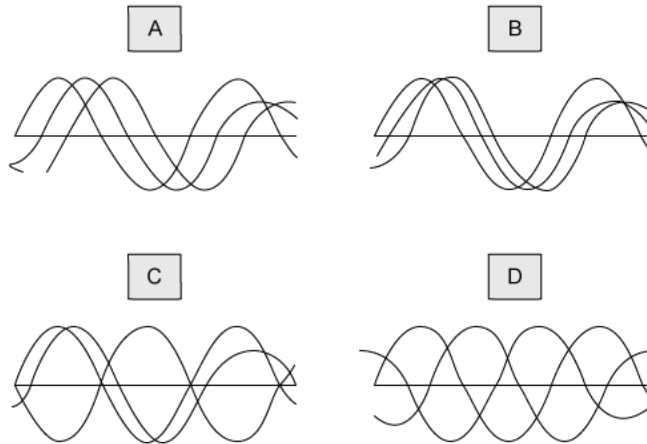
- A. 1650 Hz
- B. 830 Hz
- C. 550 Hz
- D. 413 Hz

[1 mark]

Question 6

Each of the diagrams below show three waves; each of the waves have the same amplitude and frequency but have different phases.

The waves are added together to give a resultant wave. In which case would the resultant wave be zero?



[1 mark]

Question 7

A stationary sound wave produced in a tube has a series of nodes. The distance between the first node and the sixth node is 30 cm

What is the wavelength of the wave?

- A. 12.0 cm
- B. 10.0 cm
- C. 6.0 cm
- D. 5.0 cm

[1 mark]

Question 8

A stationary sound wave produced in a musical instrument, this instrument has the pipes open at one end and closed at the other.

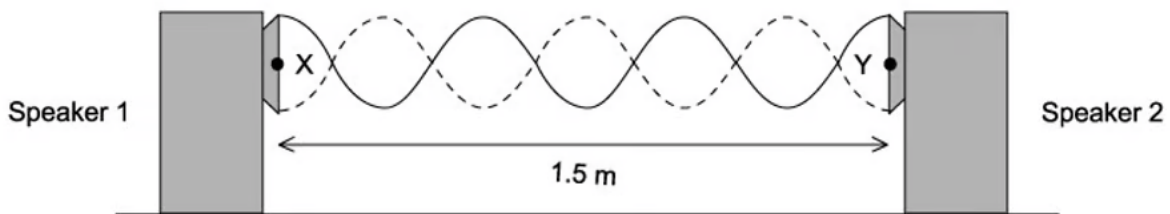
If the length of the pipe was 10 m what is the lowest frequency of sound when the speed of sound is 320 m s^{-1} ?

- A. 32 Hz
- B. 16 Hz
- C. 8 Hz
- D. 4 Hz

[1 mark]

Question 9

Two loudspeakers emitting sound of the same frequency produces a stationary wave. This is shown in the diagram



The microphone is moved between X and Y for a distance of 1.5 m. There were six nodes and seven antinodes observed.

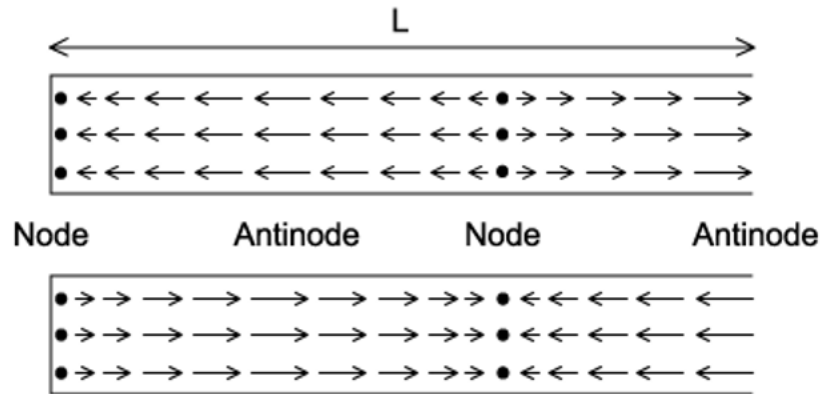
What is the wavelength of the sound?

- A. 0.21 m
- B. 0.25 m
- C. 0.43 m
- D. 0.50 m

[1 mark]

Question 10

A stationary wave was formed in a column. The diagram below shows the movement of the air particles.



The first diagram shows the displacement and the second is the displacement half a cycle later.

Which position will the pressure change by the most significant amount, and what is the length, L of the column in terms of wavelength, λ ?

	Length L	Maximum pressure change
A	$\frac{3}{4}\lambda$	antinode
B	$\frac{3}{4}\lambda$	node
C	$\frac{3}{2}\lambda$	antinode
D	$\frac{3}{2}\lambda$	node

[1 mark]



Model Answers: Medium

1

The correct answer is **A** because:

- The diagram shows one complete wave is produced with a frequency of 650 Hz
 - The wave equation is given by
- In order for the wavelength to stay the same as the frequency increases, the speed of the wave needs to be increased

B & C are incorrect because in order to maintain the same wave pattern, the wavelength must stay the same

D is incorrect because if the speed is decreased, then the wavelength will decrease changing the wave pattern

2

The correct answer is **A** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- A node is produced with zero displacement and an antinode at maximum displacement
- The distance between two nodes is half a wavelength, and there is an antinode at the middle of two successive nodes
 - Therefore, a node is a quarter of a wavelength from an adjacent antinode

B is incorrect because stationary waves do not oscillate

C is incorrect because a node would be formed at the metal sheet

D is incorrect because in a stationary wave, the points a loop are in phase and out of phase with adjacent loops

3

The correct answer is **C** because:

- The principle of superposition is applied when two or more waves are travelling through the same medium at the same time
- Noise-cancelling headphones use destructive interference to cancel out the external noise
- Therefore, the headphone will need to produce a wave with the same frequency and amplitude but with a phase difference of 180°

A is incorrect because a phase difference of 360° will mean that the waves are in phase and this will cause constructive interference making the external noise louder

B & D are incorrect because a phase difference of 90° and 270° will cause a comb filter effect, alternating constructive and destructive peaks and troughs

4

The correct answer is **A** because:

- The diagram is a transverse wave as the displacement is vertical and the direction of travel is horizontal
- 100 cm has 2.5 waves (1 loop is half a wave), so the wavelength is 40 cm
- There are nodes present that remain still, hence it is a stationary wave

B is incorrect because the diagram shows a stationary wave, not a progressive wave

C & D are incorrect because there are 2.5 waves in 100 cm, so the wavelength is 40 cm, not 20 cm

5

The correct answer is **D** because:

- The diagram shows $\frac{3}{4}$ of a wavelength is formed in a 60 cm tube
 - Therefore, one wavelength is equal to $\lambda = 80 \text{ cm} = 0.8 \text{ m}$
- The wave equation is given by $v = f \lambda$

$$\circ f = \frac{v}{\lambda} = \frac{330}{0.8} = 412.5 \text{ Hz}$$

- Therefore, the frequency of the wave is **413 Hz**

6

The correct answer is **D** because:

- In order for the displacement to be zero the sum of the displacements needs to be zero
- One wave has zero displacement, one negative and one positive, giving a resultant displacement of zero

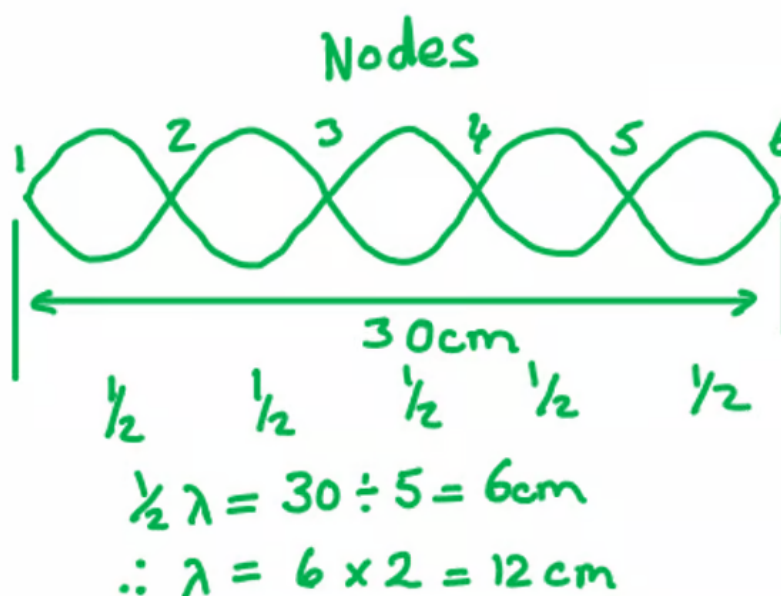
A & B are incorrect because they both give a negative displacement

C is incorrect because two of the three waves will have a negative displacement but will still leave one wave; therefore the resultant wave will not be zero

7

The correct answer is **A** because:

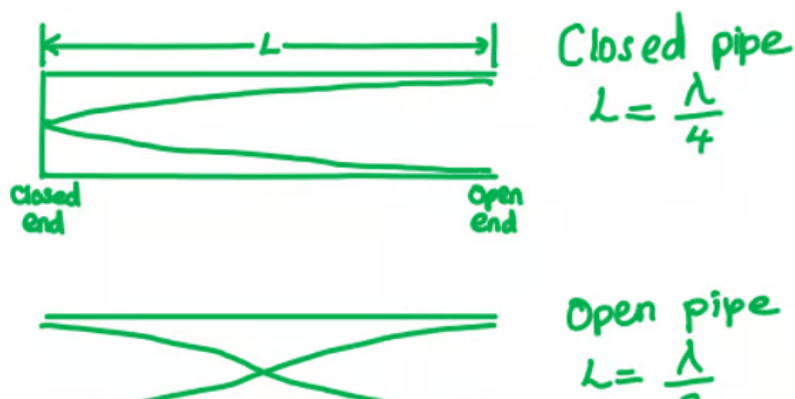
- There are five nodes in 30 cm and each have half a wavelength between each one
 - 30 cm divided by 5 = 6 cm
 - So, $\frac{\lambda}{2} = 6.0 \text{ cm}$
- Therefore, $\lambda = 12.0 \text{ cm}$



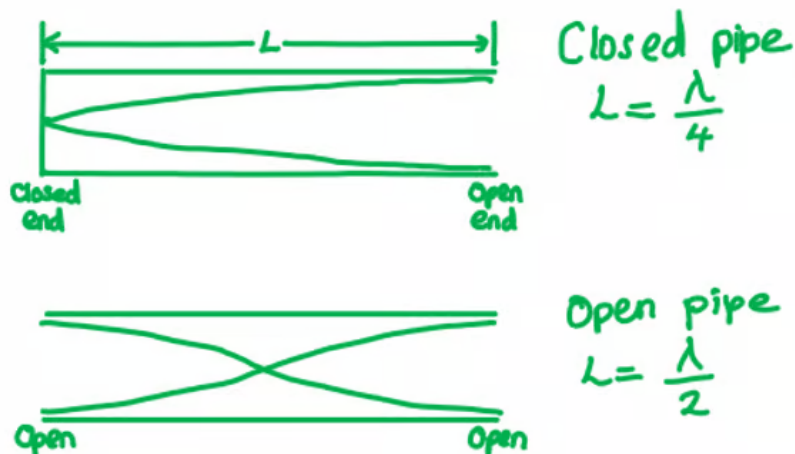
8

The correct answer is **C** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The lowest or fundamental note produced will have only one node at the closed end and an antinode at the open end, creating $\frac{1}{4}$ of a wavelength



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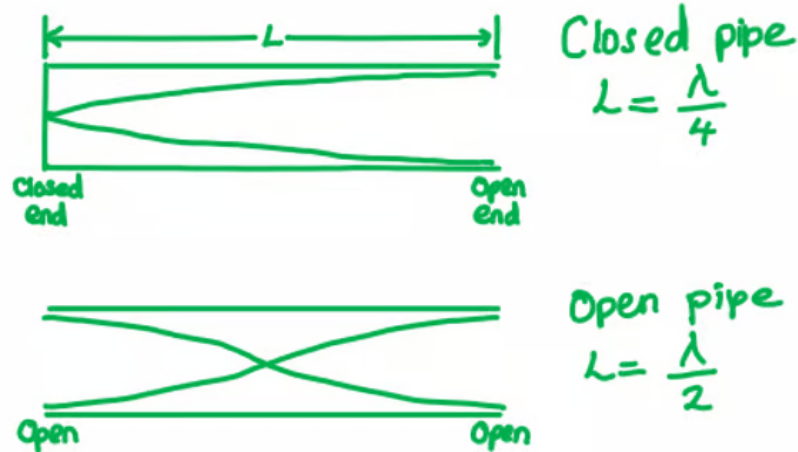


- A quarter of a wavelength is formed in a 10 m pipe
 - Therefore, one wavelength = 40 m
- The wave equation is given by $v = f \lambda$
 - $f = \frac{v}{\lambda} = \frac{320}{40} = 8 \text{ Hz}$

9

The correct answer is **D** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude



- In this open-ended pipe, there are six nodes in 1.5 m each have half a wavelength between each one
 - 1.5 m divided by 6 = 0.25 m
 - So, $\frac{\lambda}{2} = 0.25 \text{ m}$
- Therefore, $\lambda = 0.5 \text{ m}$

10

The correct answer is **A** because:

- A standing or stationary wave is formed from the superposition of two travelling waves with the same frequency, polarisation and amplitude
- The pipe is closed at one end; there are two nodes present, therefore, the length of the tube is equal to $\frac{3}{4}\lambda$

between each one

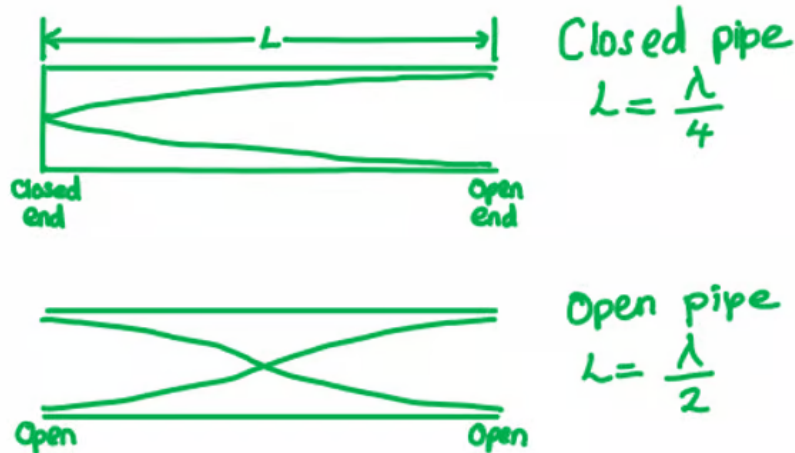
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- The maximum pressure change happens at the maximum displacement of the wave, which occurs at the antinode